



# AI-Based Customer Churn Prediction System

*Predicting customer churn using Machine Learning to help businesses retain customers*

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Amity University Online | 2025-2026

# PROJECT AGENDA

## 01 Problem Statement

Why churn prediction matters

## 02 Dataset Overview

7,043 Telco customer records

## 03 EDA Findings

Key churn patterns discovered

## 04 Model Results

LR, RF, XGBoost comparison

## 05 Dashboard

Live Streamlit dashboard

## 06 Conclusions

Findings & recommendations

# PROBLEM STATEMENT

# 26.5%

Customer Churn Rate

1,869 out of 7,043  
customers left

*"Acquiring a new customer costs  
5-7x more than retaining one"*

✗ Businesses lose revenue without knowing which customers will leave

✗ No early warning system to identify at-risk customers

✗ Reactive retention strategies applied after customer already decided to leave

✓ SOLUTION: AI model predicts churn probability BEFORE it happens

# DATASET OVERVIEW

7,043

Customer Records

21

Features

26.5%

Churn Rate

80%/20%

Train/Test Split

## Feature Categories:



### Demographics

gender, SeniorCitizen,  
Partner, Dependents



### Account Info

tenure, Contract,  
MonthlyCharges,  
TotalCharges



### Services

InternetService,  
OnlineSecurity,  
TechSupport, StreamingTV

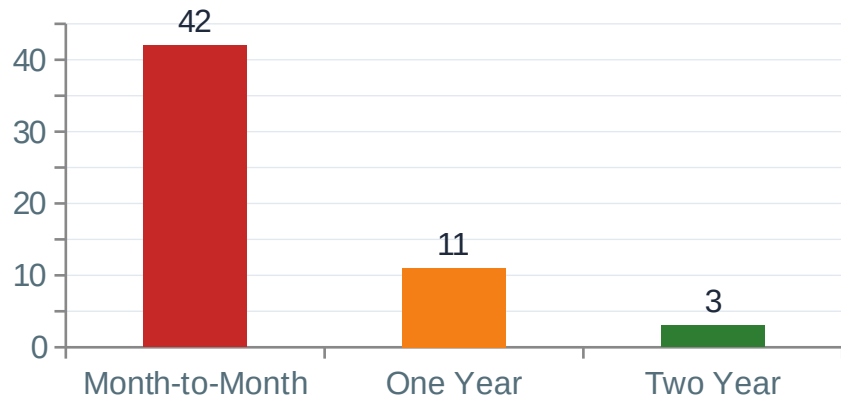


### Target

Churn (Yes/No)  
→ What we predict

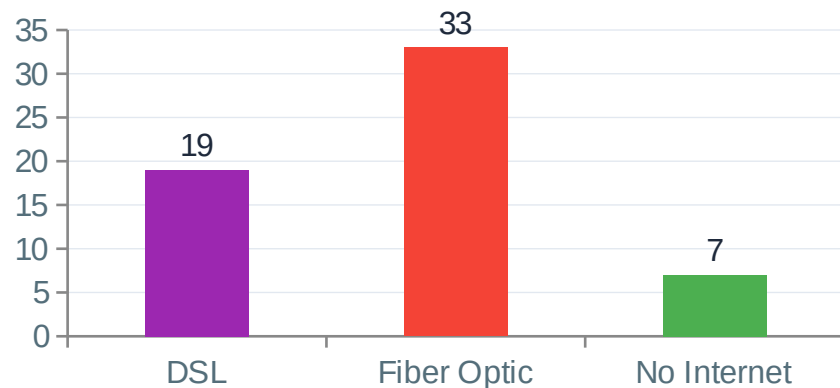
# KEY EDA FINDINGS

## Churn Rate by Contract Type



Month-to-month: 42% churn vs 3% for two-year contracts

## Churn Rate by Internet Service



New customers (0-10 months) at highest risk



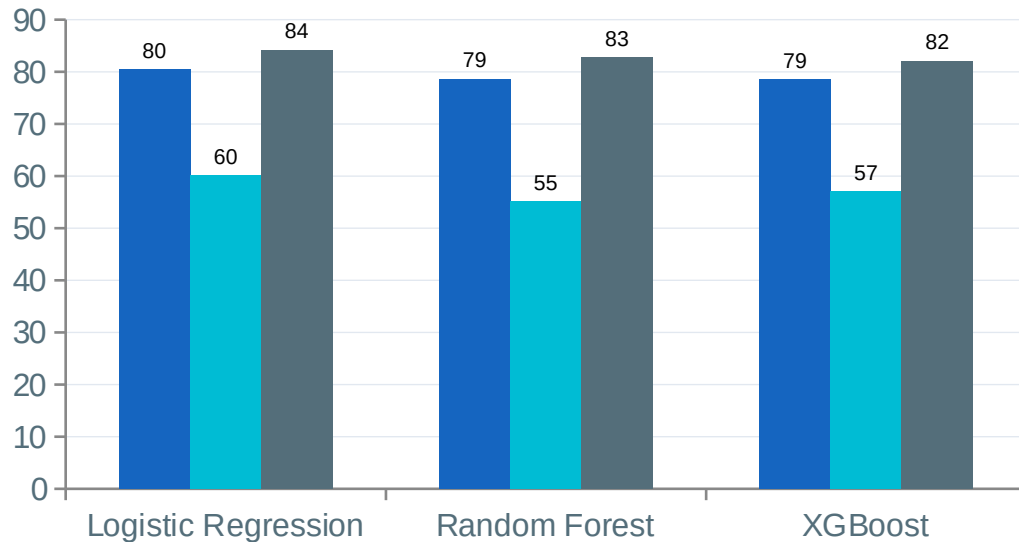
Fiber optic: 33% churn vs DSL: 19% churn



Churned customers pay higher monthly charges

# MODEL PERFORMANCE RESULTS

## Model Comparison (%)



5-Fold Cross Validation:



**BEST MODEL**

## Logistic Regression

|           |        |
|-----------|--------|
| Accuracy  | 80.48% |
| Precision | 65.71% |
| Recall    | 55.35% |
| F1-Score  | 60.09% |
| AUC       | 0.842  |
| CV Mean   | 80.43% |

# DASHBOARD DEVELOPMENT



## Customer Churn Prediction Dashboard

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7,043

Total Customers

1,869

Churned

5,174

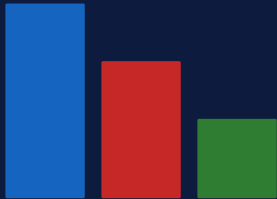
Retained

80.48%

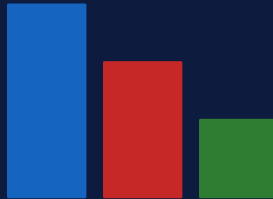
Model Accuracy



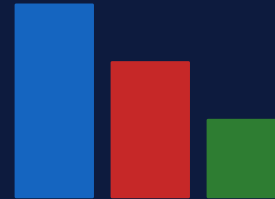
Churn Distribution



Contract vs Churn



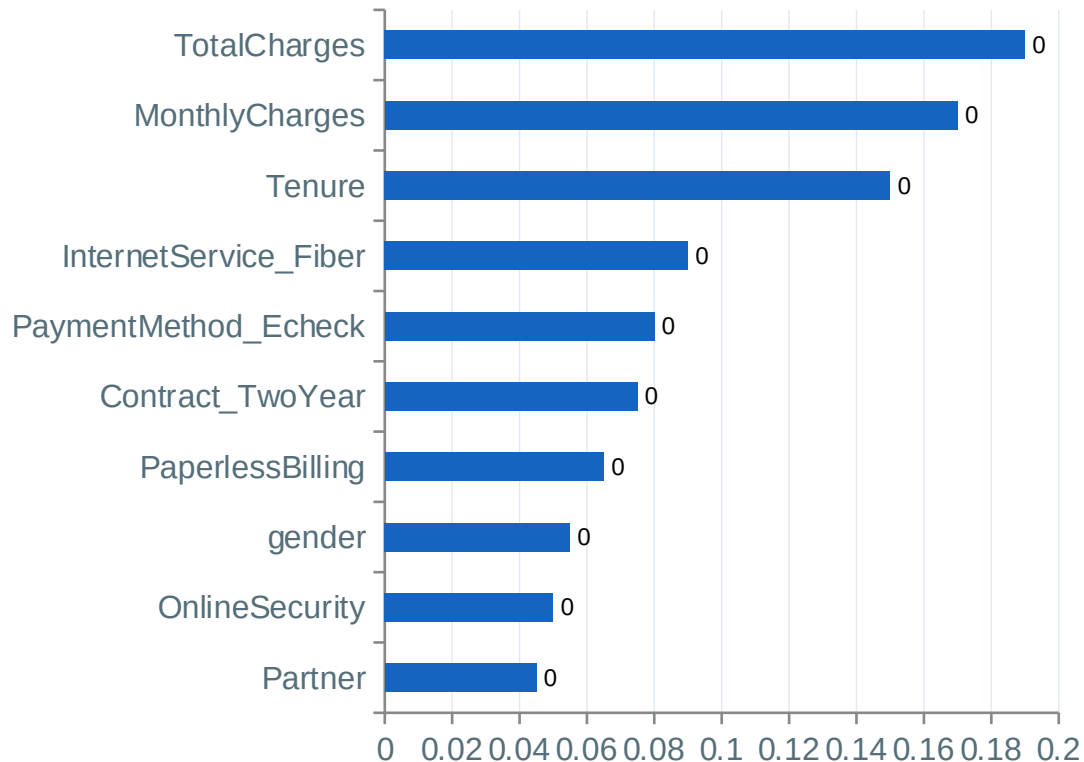
Internet vs Churn



🎯 Real-Time Churn Prediction Tool → Enter customer details → Get instant churn risk score + retention recommendations

# TOP CHURN PREDICTORS

## Random Forest Feature Importance



### #1 TotalCharges

Higher cumulative billing → higher churn risk. Long-term cost perception drives decisions.

### #2 MonthlyCharges

Customers paying premium monthly bills show greater dissatisfaction and churn tendency.

### #3 Tenure

New customers (0-10 months) churn most. Loyalty grows significantly after 24 months.

### #4 Fiber Optic

Fiber optic service users churn at 33% — possible quality or pricing dissatisfaction.



# BUSINESS RECOMMENDATIONS

## Incentivize Long Contracts

Offer 15-20% discount to move month-to-month customers to annual/2-year contracts

## Deploy AI Prediction

Run weekly churn scoring to identify at-risk customers before they leave

## Onboard New Customers

Assign dedicated managers to first 6 months customers — highest churn risk period

## Fix Fiber Optic Issues

Investigate service quality; fiber customers churn at 33% — needs urgent attention

## Loyalty Programs

Reward long-tenure customers with exclusive benefits and service upgrades

## Automate Retention

Trigger CRM workflows when churn probability exceeds 60% threshold

# PROJECT CONCLUSION

**Dataset:** 7,043 Telco customer records analyzed

**Best Model:** Logistic Regression → 80.48% Accuracy, AUC: 0.842

**EDA:** 6 visualizations revealing key churn patterns

**Dashboard:** Live Streamlit dashboard with real-time prediction

**Deployment:** Flask REST API strategy + GitHub documentation

**Key Insight:** Contract type is strongest churn predictor (42% vs 3%)

**Thank You!**

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